



- 2 > Lagrangian formalism

a) problem setup:
 let's introduce some field φ which depends on a space-time point.

notation $x \equiv (x^0 = t; x^i) \equiv (x^m)$

total dimension of space-time $(D=4)$

space coordinates $0 \leq m \leq D-1$

x belongs to some space:

$$x \in \Omega_D \subset \mathbb{R}^D$$

ex with the elastic string of length L : $D=2$

$$\Omega_2 = [t_1, t_2] \times [0, L]$$

More generally, all definition domains will look like:

$$\Omega_D = [t_1, t_2] \times \underbrace{\tilde{\Omega}_{D-1}}_{\text{spatial part}}$$

→ field $\varphi(x)$

② dynamics: we have a Lagrangian density $\mathcal{L}(\varphi, \underbrace{\partial_m \varphi}_\frac{\partial \varphi}{\partial x^m}; x)$

③ The action then reads:

$$S = \int_{\Omega_D} d^D x \mathcal{L}(\varphi, \partial_m \varphi; x)$$

Lagrangian L

Putting $c=1$ →

$$= \int_{t_1}^{t_2} dt \int_{\tilde{\Omega}_{D-1}} d^{D-1} x \mathcal{L}(\varphi, \partial_m \varphi; x)$$

• b) equation(s) of motion:

we will use the same type of reasoning:

→ Introduce a small variation of the field

→ impose that S is stationary ($\delta S = 0$)

Starting point:

$$S = \int_{\Omega_D} d^D x \mathcal{L}(\varphi(x), \partial_m \varphi; x)$$

$$\left\{ \begin{array}{l} \varphi(x) \rightarrow \varphi(x) + \delta \varphi(x) \\ \partial_m \varphi \rightarrow \partial_m \varphi + \partial_m \delta \varphi(x) \end{array} \right.$$

$$\Rightarrow S' = \int_{\Omega_D} d^D x \mathcal{L}(\varphi + \delta \varphi; \partial_m \varphi + \partial_m \delta \varphi; x)$$

$$= \int_{\Omega_D} d^D x \mathcal{L}(\varphi, \partial_m \varphi; x) \quad S$$

Taylor expansion
at first order

Einstein convention: summation over
repeated indices implied

$$\delta S + \int_{\Omega_D} d^D x \left[\frac{\partial \mathcal{L}}{\partial \varphi} \delta \varphi + \frac{\partial \mathcal{L}}{\partial (\partial_m \varphi)} \partial_m \delta \varphi \right]$$

+ higher orders

However^o

$$\frac{\partial \mathcal{L}}{\partial (\partial_m \varphi)} \partial_m \delta \varphi = \partial_m \left(\frac{\partial \mathcal{L}}{\partial (\partial_m \varphi)} \delta \varphi \right) - \left(\partial_m \frac{\partial \mathcal{L}}{\partial (\partial_m \varphi)} \right) \times \delta \varphi$$

$$\text{div } \vec{A} = \partial_m A^m$$

$$(u \cdot v') = (uv)' - v'v$$

Then^o

$$\delta S = \int_{\Omega} d^D x \left[\frac{\partial \mathcal{L}}{\partial \varphi} - \partial_m \frac{\partial \mathcal{L}}{\partial (\partial_m \varphi)} \right] \delta \varphi + \int_{\Omega} d^D x \partial_m \left(\frac{\partial \mathcal{L}}{\partial (\partial_m \varphi)} \delta \varphi \right)$$

divergence integral

Assumption: the boundary conditions on $\delta \varphi(x)$ implies that the divergence term vanishes

$$(\delta \varphi(x) = 0, \forall x \in \text{surface of } \Omega)$$

Hence:

$$\forall \delta \varphi(x), \quad \delta S = \int_{\Omega} d^D x \left[\frac{\partial \mathcal{L}}{\partial \varphi} - \partial_m \frac{\partial \mathcal{L}}{\partial (\partial_m \varphi)} \right] \delta \varphi$$

We impose $\delta S = 0$

then, because of the " $\forall$ " $\delta \varphi$,

summation over m understood

$$\frac{\partial \mathcal{L}}{\partial \varphi} - \partial_m \frac{\partial \mathcal{L}}{\partial (\partial_m \varphi)} = 0$$

Euler-Lagrange equation for a field

Comment^o: $\mathcal{L}$ and $\mathcal{L} + \underbrace{\partial_n K^n}_{\text{divergence term}}$ give the same equation of motion

• c) notion of functional :

T is a functional $\iff$ it is a function of a function

$$T : \text{function space} \longrightarrow \mathbb{R} \text{ or } \mathbb{C}$$

$$f \longrightarrow T[f] \quad \leftarrow \text{notation } []$$

We can say that a functional is the generalization of a function which depends on a set of different variables ($f(x_1, x_2, \dots)$) to a continuous set of variables

~~ex~~ $S[\varphi] \quad S'[\varphi + \delta\varphi]$

Remark : derivative of a functional

★ in the context of a multi-variable function :

$$f(x^1, x^2, \dots) \quad \underline{Pb} \quad \frac{\partial f}{\partial x^i} \quad ??$$

$$\frac{\partial f}{\partial x^i} = \lim_{\epsilon \rightarrow 0} \frac{f(x^1, x^2, \dots, x^i + \epsilon, \dots) - f(x^1, x^2, \dots, x^i, \dots)}{\epsilon}$$

$$\text{notation} = \lim_{\epsilon \rightarrow 0} \frac{f(x^j + \delta_{ij}^j \epsilon) - f(x^j)}{\epsilon}$$

$$\delta_{ij}^j : \text{Kronecker delta}$$

★ functional case:

$$f \rightarrow \Gamma \quad x^j \rightarrow \varphi(y)$$

Hence:

$$\frac{\delta \Gamma}{\delta \varphi(x)} = \lim_{\epsilon \rightarrow 0} \frac{\Gamma[\varphi(y) + \epsilon \delta^D(x-y)] - \Gamma[\varphi(y)]}{\epsilon}$$

$\delta^D(x) = D$ -dimensional Dirac "function"

Note: remember that

$$\delta S = \int_{\Sigma_D} d^D x \left(\frac{\partial \mathcal{L}}{\partial \varphi} - \partial_\mu \frac{\partial \mathcal{L}}{\partial (\partial_\mu \varphi)} \right) \delta \varphi$$

definition $\rightarrow$ $\equiv \int_{\Sigma_D} d^D x \frac{\delta S}{\delta \varphi(x)} \delta \varphi(x)$

$$\frac{\delta S}{\delta \varphi(x)}$$

$$= \frac{\delta S}{\delta \varphi(x)} = \frac{\delta \mathcal{L}}{\delta \varphi(x)} - \partial_\mu \frac{\partial \mathcal{L}}{\partial (\partial_\mu \varphi(x))}$$

Exercise

compute $\frac{\delta \Gamma}{\delta \varphi(x)}$

for

- $\Gamma[\varphi] = \frac{1}{2} \int_{\Sigma_D} d^D x \varphi(x)^2$
- $\Gamma[\varphi] = \varphi(y)$
- $\Gamma[\varphi] = \frac{1}{2} \int_{\Sigma_D} d^D x d^D y \varphi(x) G(x, y) \varphi(y)$

with $G(x, y) = G(y, x)$

→ Hamiltonian formalism

- a) discrete case

let's go back to the elastic string

Starting point : Lagrangian $L = a \sum_i L_i = a \sum_i \left(\frac{1}{2} \frac{M}{a} \dot{\varphi}_i^2 - \frac{1}{2} k \left(\frac{\varphi_{i+1} - \varphi_i}{a} \right)^2 \right)$

By definition, the conjugated momentum of φ_i is

$$\pi_i \stackrel{\text{def}}{=} \frac{\partial L}{\partial \dot{\varphi}_i} = a \frac{\partial L_i}{\partial \dot{\varphi}_i}$$

Then the hamiltonian is

$$H = \left(\sum_i \pi_i \dot{\varphi}_i \right) - L = a \sum_i \left(\frac{\pi_i}{a} \dot{\varphi}_i - L_i \right)$$

- b) Continuous limit :

$$L_i \rightarrow \mathcal{L}$$

$$L \rightarrow \int_{\tilde{\Sigma}_{D-1}} d^{D-1}x \mathcal{L}$$

$$a \sum_i \rightarrow \int_{\tilde{\Sigma}_{D-1}} d^{D-1}x \quad (\text{space-only})$$

$$\frac{\pi_i}{a} = \frac{\partial L_i}{\partial \dot{\varphi}_i} \longrightarrow \boxed{\pi(x) = \frac{\partial \mathcal{L}}{\partial (\partial_t \varphi)}} \quad \left(\partial_t = \frac{\partial}{\partial t} \right)$$

conjugate
momentum
density
(FIELD)

Finally

$$H = \int_{\Sigma_{D-1}} d^{D-1}x \left[\pi(x) \dot{\varphi}(x) - \mathcal{L}(\varphi, \partial_m \varphi, x) \right]$$

hamiltonian density $\mathcal{H}$

Comment:

- Lagrangian $\rightarrow S$ (time & space are on the same foot)
- Hamiltonian $\rightarrow H$ (time is singled out)

$\rightarrow$ Symmetries and Noether's Theorem

- a) Problem set-up

Let's start with a field $\varphi(x)$ whose dynamics is driven by $\mathcal{L}(\varphi, \partial_m \varphi; x)$

$$S[\varphi] = \int_{\Sigma_D} d^Dx \mathcal{L}(\varphi, \partial_m \varphi; x)$$

There are two types of transformations which can modify the fields:

① Continuous transformation:

They can be infinitesimal, local, Non-local

ex: $t \rightarrow t' = t + a$

$|a| \ll 1$ infinitesimal transf.

$a = ct$ Non-local transf.

$a(x)$: local //

Note: Transformations which act upon the coordinates are geometrical transf.

Transformations which act upon the field density are called internal transf.

② Discrete transformations

Those ones cannot be infinitesimal

ex $t \rightarrow -t$

From now on, we will use continuous transformations

i.e. we will use a transformation which can act on the field and its coordinates

$$x^M \rightarrow x'^M = x^M + \delta x^M \ll 1$$

$$\partial_M = \frac{\partial}{\partial x^M} \quad \psi(x) \rightarrow \psi'(x')$$

↖ $\partial_M \psi(x) \rightarrow \partial'_M \psi'(x')$

• b) Symmetry

Definition: the transformation is a symmetry $\iff$ the action after the transformation is equal to the action before $\iff S[\varphi] = S'[\varphi']$

$$\iff \int_{x_0}^{x_1} d^0 x' \underbrace{\mathcal{L}'(\varphi', \partial'_m \varphi'; x')}_{S'[\varphi]} = \int_{x_0}^{x_1} d^0 x \underbrace{\mathcal{L}(\varphi, \partial_m \varphi; x)}_{S[\varphi]}$$

Lograngian density after the transformation

Remark: a symmetry gives another possible motion of a system

ASSUMPTION $\rightarrow$ the invariance of the action may result from different relationships between $\mathcal{L}$ and $\mathcal{L}'$

The two most natural possibilities are:

① both Lagrangian have the same mathematical structure

$$\mathcal{L}'(\varphi', \partial'_m \varphi'; x') = \mathcal{L}(\varphi', \partial'_m \varphi'; x')$$

② They differ by a divergent term

$$\mathcal{L}'(\varphi', \partial'_m \varphi'; x') = \mathcal{L}(\varphi', \partial'_m \varphi'; x') + \partial'_n K^n(\varphi', \partial'_m \varphi'; x')$$

However, for all the transformations encountered in these lectures and almost in real life (no counter example found yet); the #2 case never occurs.

CONSEQUENCE

$$S'[\varphi'] = S[\varphi]$$

$$\Leftrightarrow \int_{\Omega_0} d^D x \mathcal{L}(\varphi, \partial_M \varphi, x) = \int_{\Omega'_0} d^D x' \mathcal{L}'(\varphi', \partial_M \varphi'; x')$$

change of variable $x' \rightarrow x$

$$= \int_{\Omega_0} d^D x \underbrace{\left| \frac{\partial x'}{\partial x} \right|}_{\substack{\text{Determined} \\ \text{of Jacobian} \\ \text{matrix}}} \underbrace{\mathcal{L}'(\varphi', \partial_M \varphi'; x')}_{\substack{\text{here, functions} \\ \text{of } \varphi', \partial_M \varphi', x}}$$

By identification

$$\mathcal{L}(\varphi, \partial_M \varphi; x) = \left| \frac{\partial x'}{\partial x} \right| \mathcal{L}'(\varphi', \partial_M \varphi'; x')$$

Using the assumption on $\mathcal{L}$ and $\mathcal{L}'$

$$\rightarrow = \left| \frac{\partial x'}{\partial x} \right| \mathcal{L}(\varphi', \partial_M \varphi'; x')$$

Pb Perform a Taylor expansion at first order of " $S[\varphi] = S'[\varphi]$ " to get the Noether theorem.